



DUMMY EXAM

COMS3005A

Advanced Analysis of Algorithms

Duration 90 minutes

This paper consists of 3 pages.

Instructions

1. Answer all questions.
2. The mark for each question is given in brackets next to each question.
3. Read each question carefully before answering: always make sure you provide an answer to what is being asked.

[TURN PAGE]

Question 1	Greedy Algorithms	[4]
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- (a) Let $A_1, \dots, A_n$ be matrices where the dimensions of A_i are $d_{i-1} \times d_i$ for $i = 1, \dots, n$.
A suggested greedy algorithm to determine the best order in which to perform the matrix multiplications to compute $A_1 \times A_2 \times \dots \times A_n$ is to choose the largest remaining dimension from $d_1 \dots d_{n-1}$ and multiply two adjacent matrices that share that dimension.
Will the algorithm always give an optimal number of multiplications?
Explain your answer. (4)

Question 2	Dynamic Programming	[20]
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- (a) Given a binary tree containing n nodes, derive a dynamic programming algorithm to decide whether the tree contains a path from the root to a leaf of length k , $k \leq n$.
Discuss the efficiency of your solution. (10)
- (b) Suppose you are asked solve the problem given below.

Given a bit string (a string of 0s and 1s), determine whether or not there is a string of 1s of length K for some integer K .
Derive a dynamic programming algorithm to solve the problem. (10)

Question 3	Computational Geometry	[14]
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- (a) Suppose you are given a set of points in the plane and you know in advance that most of the points ($> 90\%$ of them) lie in a small cluster and the remaining points are scattered a long way from the cluster.
Explain what effect this situation would have on the running of the final version of the closest pair of points algorithm. (4)
- (b) Suppose you are given a set of line segments in the plane.
Each line segment is defined by its endpoints and each endpoint is defined by its x and y coordinate pairs.
- i. Derive a brute force algorithm to find the longest line segment in the plane. (3)
 - ii. Now derive an improved algorithm to find the longest line segment. (7)

Question 4	Hashing	[8]
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- (a) Suppose that you are designing a system for a golf club and you want to store the information about the members of the club in a hash table to facilitate quick look up.
One of your colleagues suggests that you use an open address hashing system where the golfer's handicap is used as the *hash code* and that you use *linear probing* to resolve collisions.
You know that the club has 620 members which means that your hash table has to have at least that many slots in order to store the information about all the members.

You also know that handicaps range between 0 and 36 and that the majority (around 70%) of golfers have handicaps between 22 and 26 (inclusive).

Would your colleague's suggestion be an effective approach to follow?

Justify your answer.

(3)

- (b) Suppose that you want to store your friends' telephone numbers in a hash table.

You have fewer than 100 friends and acquaintances so you choose 100 as the size of your hash table.

You decide to use *closed address* hashing.

Suggest a hashing function that you could use in this application and discuss whether it is likely to be a good hash function.

(5)

Question 5	Red Black Trees	[12]
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- (a) Red-black trees and hash tables can both be considered "dictionaries" in that they support fast lookup.

Describe under what circumstances one would use a hash table in preference to a red-black tree and vice versa.

(4)

- (b) Draw all of the valid red black trees with 3 nodes (excluding Nil nodes).

(4)

- (c) Draw a red black tree with 5 nodes and a black height of 2.

(2)

- (d) Now add a node to the longest branch in the tree using the standard insertion algorithm.

Will this cause a violation of any of the red black tree properties?

Explain your answer.

(2)

Question 6	General	[12]
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- (a) Design a *divide and conquer* algorithm which will take as input an unsorted list of numbers and will return two lists of numbers – one containing all of the negative numbers and one containing all of the non-negative numbers from the original list.

(4)

- (b) The algorithm below finds the position of a user defined number (x) in a list of numbers (L).

ALGORITHM DoIt

Input x

Index := 1

While Index <= n and L[Index] <> x do

 index := index + 1

If Index > n Then Index := 0

Output Index

Perform the best, average and worst case analysis on this algorithm under the assumption that the number being searched for is always in the list.

Now perform the analysis under the assumption that the number being searched for might not be in the list.

(8)

Total: 70